

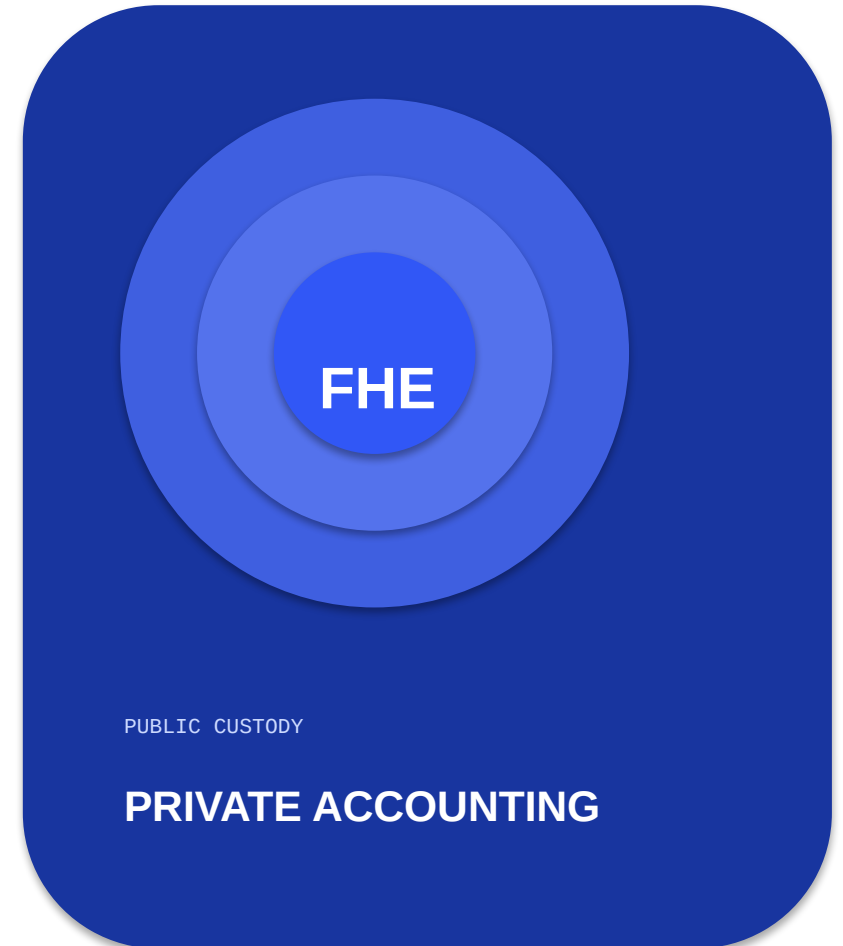
ZAMA FHEVM · CONFIDENTIAL FINANCE

Private prize savings, built for verification.

Principal stays withdrawable while encrypted balances and ticket weights stay confidential.

cipherpool-beta.vercel.app

RESEARCH SOFTWARE · NOT PRODUCTION



THE PROBLEM

Public ledgers expose private financial context

Auditability should not require publishing every saver's position.

01

Balance exposure

Wallet history can reveal a saver's position, deposits, withdrawals, and changing financial behavior.

02

Visible ticket weights

Transparent weights make whale activity easy to track and can disclose each participant's relative odds.

03

Timing reveals intent

Pending deposits and last-minute entries expose behavior around a prize round before it settles.

THE SOLUTION

Public custody. Confidential accounting.

CipherPool separates what must be auditable from what should remain private.

PUBLIC · INDEPENDENTLY VERIFIABLE

Custody layer

- Deposited testnet cUSDC
- Contract and custody bindings
- Transaction receipts
- Aggregate solvency

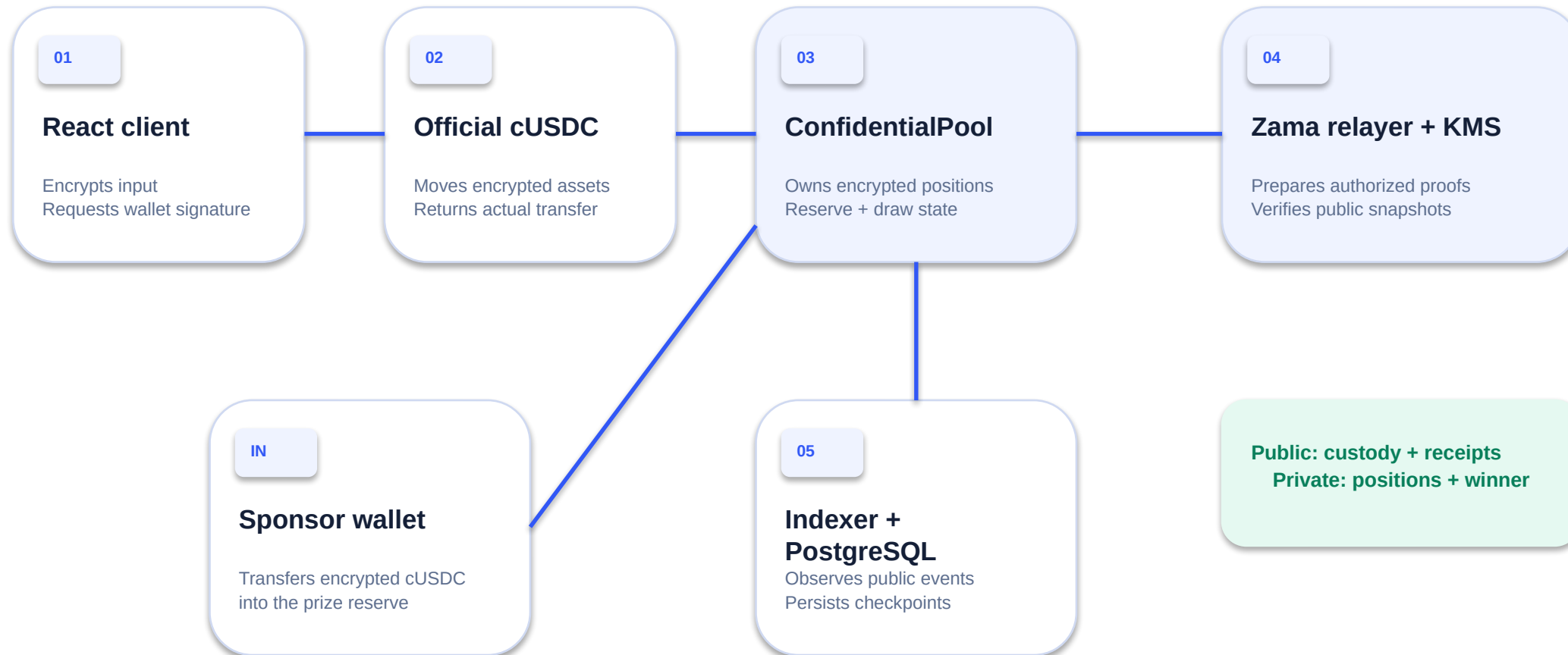
ENCRYPTED · WALLET-AUTHORIZED

Privacy layer

- Running balances as euint64
- Weighted draw intervals
- Prize credits
- Local balance reveal

ARCHITECTURE

A verifiable path from wallet to confidential settlement



LIVE PRODUCT

The console is live—and honest about connection state

The screenshot displays the CipherPool Confidential Savings Console. At the top, the header includes the CipherPool logo, navigation links for 'Save privately', 'Prize rounds', and 'How it works', and a 'Connect wallet' button. The main content area is titled 'CONFIDENTIAL SAVINGS CONSOLE' and features a large heading 'Your position stays private.' with a subtext explaining the confidentiality of deposits and withdrawals. A 'RUNTIME ASSURANCE' section shows the 'Protocol state' as 'PENDING', with indicators for 'Network' (Wallet Disconnected), 'Indexer' (Online), and 'Transactions' (Safety-Locked). A yellow warning banner states: 'Writes remain locked until runtime verification succeeds. Connect a Sepolia wallet to verify the active deployment.' Below this, a status bar shows 'Indexer Online', 'Pool Checking', 'Wallet Disconnected', and 'Deployment Pending'. The console is divided into four main sections: 'VERIFIED POOL SNAPSHOT' (UNAVAILABLE), 'VERIFIED PRIZE RESERVE' (UNAVAILABLE), 'PRIVATE SAVERS' (LOADING), and 'CONFIRMED ROUNDS' (LIVE). The bottom section, 'PRIVATE OPERATIONS', is titled 'Manage your savings' and includes three steps: 'STEP 01 Fund your private position' (ENCRIPTED END-TO-END) and 'STEP 02 Withdraw confidentially' (ONE TRANSACTION). A note at the bottom right states: 'Balances reveal only after wallet authorization. Deposits and settlement remain independently verifiable on-chain.'

PRODUCTION CAPTURE

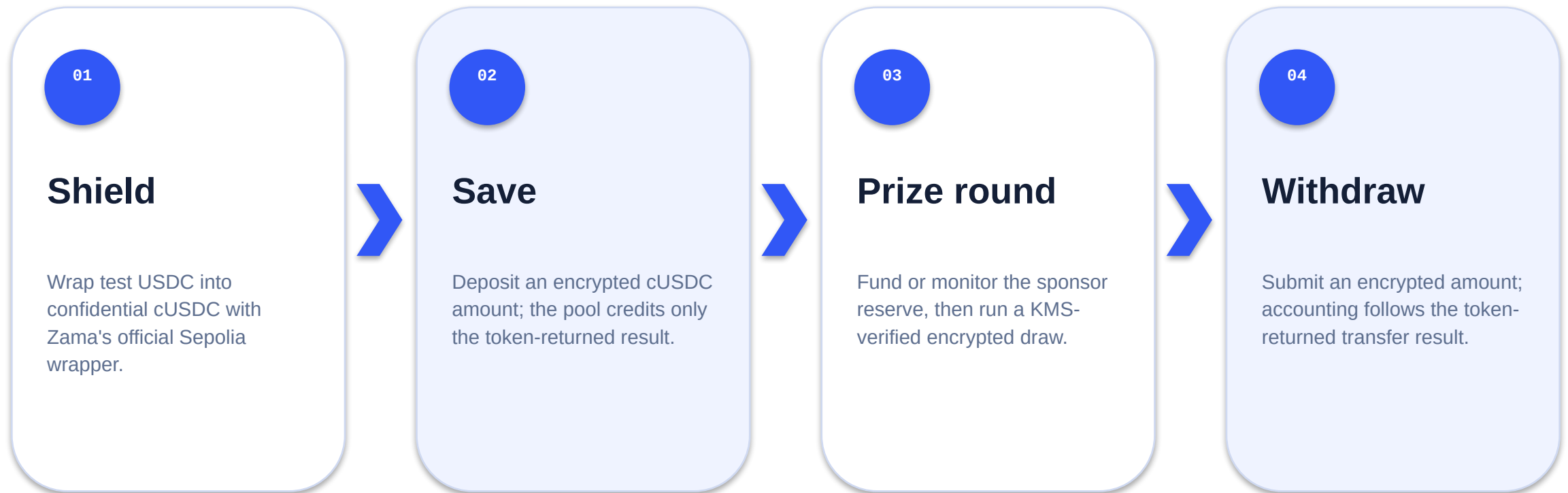
Clear wallet state

A disconnected user sees only Connect wallet. An address appears only after the provider returns the selected account.

Runtime assurance

Chain, bytecode, and custody bindings are checked before protocol writes.

Four explicit steps. No hidden signing.



SECURITY CONTROLS

Solvency and settlement invariants are enforced in code

01

Asset-bound credit

Encrypted deposit credit is derived from the token-returned custody amount.

02

Reserved prizes

Each draw consumes verified sponsor funds so assets cannot fund repeated awards.

03

Complete liabilities

Winner credit enters both the encrypted position and aggregate at finalization.

04

Bound draw proofs

KMS evidence must match the stored aggregate and reserve handles.

05

Timeout recovery

Anyone can release a stale draw lock after the 24-hour cancellation delay.

VERIFIED EVIDENCE

A real 1 cUSDC Sepolia lifecycle—not a simulated receipt

Each transaction below links directly to its confirmed block-explorer record.

01

cUSDC deposit

[0x36f81f...fa87](#)

Encrypted position

02

Direct withdrawal

[0x8ee0e4...8429](#)

Wallet restored

03

Sponsor reserve

[0x07b797...7eaa](#)

Encrypted prize funds

Round trip restored 10 cUSDC · later sponsor funding moved 1 cUSDC into reserve

Reproducible tests—and durable live infrastructure

FULL

VALIDATION SUITE

Foundry contracts · backend API/indexer
client encryption · frontend UX

Vercel

React frontend

PRODUCTION UI

Render

Node indexer/API

PUBLIC API

Neon

PostgreSQL checkpoint

RESTART RECOVERY

Indexer checkpoint restored successfully after a live service restart

OPEN RESEARCH · VERIFIABLE EVIDENCE

Confidential savings, without blind trust.

Try the live app. Inspect the active Sepolia deployment. Review every invariant and test.

cipherpool-beta.vercel.app

github.com/Webghost01-NG/fhevm-pooltogether-security

ACTIVE POOL

**0xE47eF44E
BB804A50
7173BEFa
5beb2325
aA7451AD**

ETHEREUM SEPOLIA